Hot carrier-dynamics

Hot carrier-dynamics in Silicon single crystals

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Motivation & Background

- **Shockley-Quaser:** Thermodynamical limiting efficiency for solar panels.
- **Hot carrier-dynamics:** Very little known about hot state conductivity.
- **Photocurrent spectroscopy:** Measuring carrier-dynamics with pico-second resolution.

Figure:

Photo- current/ conductivity Spectroscopy

- Optical pump electrical probe
- Electrical detection
- Advantages over TA methods

Figure:

Experimental Methods

Conductivity changes

The material responds to illumination by changing its conductivity:

$$\Delta\sigma = \Delta\mu\Delta N \tag{1}$$

Key Properties

- $\Delta\sigma = \Delta\sigma_1 + \Delta\sigma_2 + C(\tau)$
- μ is related to carrier-dynamics.
- N is related to carrier-population.

Important Consideration

The different states have their own carrier mobility μ_i.

Experimental Setup

Figure:

Carrier Two-Temperature Model

Governing Equations

Carrier density model

$$\Delta\sigma = \Delta\mu\Delta N \tag{2}$$

Carrier temperature model

$$\Delta\sigma = \Delta\mu\Delta N \tag{3}$$

Lattice temperature model

$$\Delta\sigma = \Delta\mu\Delta N \tag{4}$$

Drude conductivity

$$\sigma = \frac{\sigma_0}{\nu\left(T_l\right)} \tag{5}$$

Carrier Two-Temperature Model

Figure:

Electrical Detection

The issue with electrical detection

Figure:

The solution

Figure:

Electrical Detection

Box car

$$A[k] = \frac{1}{N} \sum_{i=0}^{N-1} \left(Signal\,[i] - Background\,[i] \right). \tag{6}$$

BOX CAR AVERAGER

Electrical Detection

Box car arm

Figure:

Component	**Bandwidth [GHz]**	**Noise [$V_{rms}/\sqrt{Hz}$]**
Pre-amp	8	0.342
Gated integrator	86.15	0.281
ADC	88.38	0.234

Electrical Detection

Box car trigger

Figure:

Trigger stage	Imparted delay	Phase jitter [ppm/Hz]
Rough delay	1 ms	0.342
Fine delay	100 ns	0.281
Trigger generation	500 ps	0.234

Electrical Detection

SNR formula

The formula for signal-to-noise ratio assuming *abc*

$$SNR = someformula \tag{7}$$

Figure:

Different Photo- current/ conductivity Techniques

- Correlation spectroscopy
- Vibrationally constrolled spectroscopy
- Near field
- ...?